

MES Current-State Assessment & Target-State Blueprint

Formulas assessment and target-state blueprint for fictional regulated manufacturing

Executive summary

This assessment frames a fictional life sciences manufacturing environment where MES execution, SOP access, QA review, QC/LIMS visibility, scheduling handoffs, automation context, and process data are not yet operating as one coherent execution system.

Objective and scope

- Assess current-state friction across batch execution, QA readiness, QC/LIMS status, scheduling, MES engineering, and automation/IT-OT handoffs.
- Define a target-state operating model for MES-enabled execution and review readiness.
- Show fictional, non-proprietary examples only; not a validated site assessment.

Current-state risks

- Execution variability caused by inconsistent prompts, workarounds, and exception paths.
- Fragmented SOP/reference access at point of execution.
- Disconnected QC results and sample readiness signals.
- Manual handoffs across scheduling, QA, and operations.
- Delayed QA readiness and limited batch review visibility.

Pain-Point Matrix by Function

Function-level assessment view

Function	Observed pain point	Risk / impact
Manufacturing	Execution prompts vary by product/line; operators rely on tribal knowledge.	Higher correction effort and inconsistent exception documentation.
QA	Review package completeness is discovered late.	Delayed readiness, manual reconciliation, and avoidable rework.
QC/LIMS	Sample status and results are not visible in the execution context.	Release decisions require extra follow-up and status chasing.
Scheduling	Schedule changes do not always align with batch/system readiness.	Priority conflicts and unclear readiness signals.
MES Engineering	Recipe support queues mix defects, changes, and how-to questions.	Support demand is reactive and hard to prioritize.
Automation/IT-OT	Process context is available but not always connected to review needs.	Automation data is underused for batch context and investigation.

Target-State Operating Model

Future-state system flow and accountability

Target-state system flow

ERP -> MES -> Automation -> LIMS -> Process Data -> QA Review

ERP releases demand and material context. MES orchestrates recipe execution, checks, signatures, and batch record status. Automation and process data provide execution context. LIMS contributes sample status/results.

QA review receives structured exceptions, evidence, and release readiness signals.

Target-state operating model

- MES is the operational system of record for batch execution state and exception context.
- Operators, QA, QC, scheduling, MES engineering, and automation support share a common readiness model.
- Review readiness is visible earlier through structured batch status, missing item, deviation, and sample indicators.
- SOP/reference access is embedded near execution steps, reducing search time and informal workarounds.

Gap Assessment and Recommendations

Prioritized assessment output

Current state	Risk / impact	Target state	Priority
SOP access: Fragmented references	Search time, wrong-version risk	Step-level controlled references	High
QC status: Disconnected sample visibility	Release follow-up and delays	MES/LIMS readiness signal	High
QA review: Completeness known late	Manual reconciliation	Review readiness dashboard	High
Exception routing: Inconsistent capture	Deviation effort and rework	Standard exception pathways	Med
Scheduling handoff: Readiness not shared	Priority churn	Shared batch readiness model	Med

Recommendations and success measures

- Standardize MES execution prompts, exception capture, and review readiness indicators.
- Integrate QC/LIMS and process data status into batch context where appropriate.
- Measure review readiness, exception cycle time, right-first-time execution, and schedule adherence.